F18CD Mark Scheme - Assignment 2

1. (a)

Proof. Let (s_n) denote the sequence of partial sums for $\sum_{k=1}^{\infty} a_k$ and (t_n) denote the sequence of partial sums for $\sum_{k=1}^{\infty} b_k$. Then since $0 \leq a_k \leq b_k$ for all $k \in \mathbb{N}$ it follows that (s_n) is monotone increasing. Additionally, $s_n \leq t_n$ for all $n \in \mathbb{N}$. If $\lim(t_n) = t$, it follows by the order limit theorem that $s_n \leq t$ for all $n \in \mathbb{N}$. Hence, by the monotone convergence theorem (s_n) is convergent. □

(b)

Proof. Suppose that $\sum_{k=1}^{\infty} a_k$ converges absolutely. Then $\sum_{k=1}^{\infty} a_k$ converges by the absolute convergence test. Hence, there exists an $N \in \mathbb{N}$ in which $|a_n| < 1$ for all $n > N$. Thus, for all $n > N$ it follows that $|a_n^2| \leq |a_n| < 1$. It follows then by the comparison test that $\sum_{N+1}^{\infty} |a_n^2|$ converges. Finally, as

$$\sum_{k=1}^{\infty} |a_n^2| = \sum_{k=1}^{N} |a_n^2| + \sum_{k=N+1}^{\infty} |a_k^2|$$

it follows that $\sum_{k=1}^{\infty} |a_n^2|$ is convergent and hence $\sum_{k=1}^{\infty} a_n^2$ is absolutely convergent. □

(c) This is false. A counter-example would be let $a_k = (-1)^k/k$. Then clearly

$$\sum_{k=1}^{\infty} |a_k^2| = \sum_{k=1}^{\infty} \frac{1}{k^2}$$

which converges by the P-series test, yet

$$\sum_{k=1}^{\infty} |a_k| = \sum_{k=1}^{\infty} \frac{1}{k}$$

diverges since it is the harmonic series.

2. (a)

Proof. Let

$$a_k = \frac{1}{k(\ln(1+k))^m}.$$

Assume that $m > 0$. Since $\ln(k) \leq 0$ for all $k \in \mathbb{N}$ it follows that $a_k \leq 0$ for all $k \in \mathbb{N}$. Define the function $f : (0, \infty) \to \mathbb{R}$ given by

$$f(x) = \frac{1}{x \ln(1+x)^m}$$

with $m \in (0, \infty)$. It follows then that

$$\frac{\mathrm{d}}{\mathrm{d}x} \left(\ln(f(x))\right) = \frac{f'(x)}{f(x)} = -\frac{1}{x} - \frac{m}{(1+x)\ln(1+x)}.$$

Since $x > 0$ and $m > 0$ it follows that

$$\frac{f'(x)}{f(x)} < 0$$

for all $x \in (0, \infty)$. Clearly $f(x) > 0$ for all $x \in (0, \infty)$ and thus $f(x)$ is monotone decreasing on $(0, \infty)$. It follows that (a_k) is monotone decreasing for $m > 0$. Applying the Cauchy condensation test gives $\sum_{k=1}^{\infty} a_k$ converges if and only if $\sum_{k=1}^{\infty} 2^k a_{2^k}$ converges. The terms $2^k a_{2^k}$ are given by

$$2^k a_{2^k} = \frac{1}{\ln(1 + 2^k)^m}.$$

Now, since $2^k \leq 2^k + 1 \leq 2^{k+1}$ it follows that

$$\frac{1}{(k+1)^m \ln(2)^m} \leq \frac{1}{(\ln(1 + 2^k))^m} \leq \frac{1}{k^m \ln(2)^m}.$$

Hence, by the comparison test and P-series test, it follows that $\sum_{k=1}^{\infty} a_k$ diverges if $m \in (0, 1]$ and converges if $m > 1$. Finally, if $m = 0$ the series clearly diverges as it is the harmonic series and if $m < 0$, then $a_k > 1/k$ for all $k \in \mathbb{N} \setminus \{1\}$ and hence by the comparison test $\sum_{k=2}^{\infty} a_k$ diverges. Hence $\sum_{k=1}^{\infty} a_k$ converges if and only if $m > 1$. □

(b)

Proof. Define

$$b_k = \frac{1}{F_k F_{k+2}}.$$

Let

$$\frac{1}{F_k F_{k+2}} = \frac{A}{F_k} + \frac{B}{F_{k+2}}$$

for some $A, B \in \mathbb{R}$. Setting $A = 1/F_{k+1}$ and $B = -1/F_{k+1}$ gives the desired result. Hence, the sequence of partial sums for $\sum_{k=1}^{\infty} a_k$ is given by

$$s_n = \frac{1}{F_1 F_2} - \frac{1}{F_{n+1} F_{n+2}}.$$

Since $\lim(F_k) = \infty$ it follows that $\lim(s_n) = 1$. □

(c)

Proof. Define

$$a_k = \frac{1}{F_k F_{k+4}}.$$

Let

$$\frac{1}{F_k F_{k+4}} = \frac{A}{F_k} + \frac{B}{F_{k+4}}$$

for some $A, B \in \mathbb{R}$. Setting $A = 1/3F_{k+2}$ and $B = 1/3F_{k+2}$ gives the desired result. Chaining this with part (b) gives

$$\frac{1}{F_k F_{k+4}} = \frac{1}{3}\left(\frac{1}{F_k F_{k+1}} - \frac{1}{F_{k+1}F_{k+2}} + \frac{1}{F_{k+2}F_{k+3}} - \frac{1}{F_{k+3}F_{k+4}}\right).$$

Hence, the sequence of partial sums of $\sum_{k=1}^{\infty} b_k$ is given by

$$t_n = \frac{1}{3}\left(\frac{1}{F_1 F_2} - \frac{1}{F_{n+1}F_{n+2}} + \frac{1}{F_3 F_4} - \frac{1}{F_{n+3}F_{n+4}}\right).$$

Finally, it follows that $\lim(t_n) = 7/18$. □